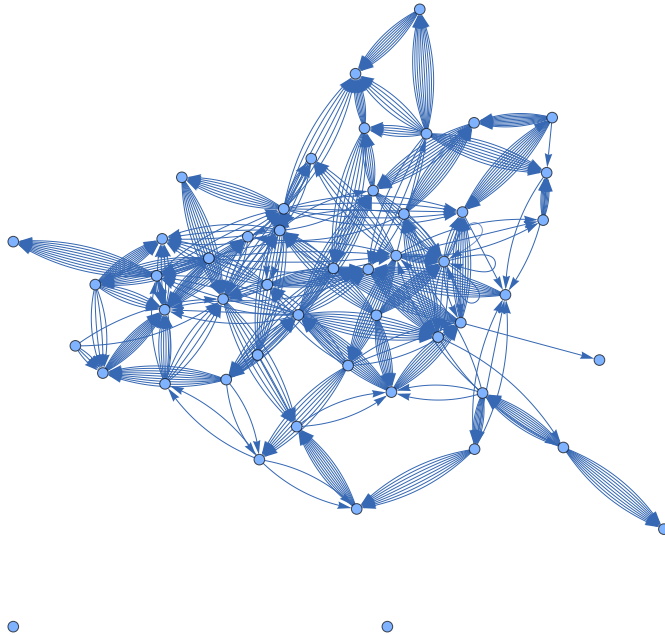


```
In[105]:=
M = With[{n = 50}, Table[If[RandomReal[] < .04, RandomInteger[{1, 10}], 0], n, n]];
```

```
In[123]:=
G = AdjacencyGraph[M]
```

```
Out[123]=
```



```
In[124]:=
totals = Total[M, {2}];
Ordering[totals]
```

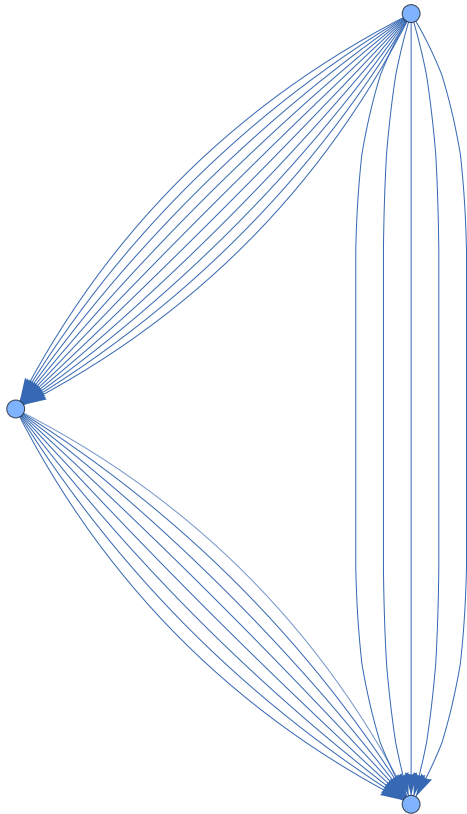
```
Out[125]=
{1, 15, 26, 33, 39, 42, 50, 25, 38, 6, 24, 36, 49, 8, 27, 4, 10, 32, 9, 23, 30, 40, 43, 46, 48,
 19, 2, 11, 31, 35, 45, 3, 20, 18, 28, 44, 5, 13, 21, 29, 37, 41, 17, 47, 12, 14, 22, 7, 16, 34}
```

```
In[126]:=
gi_ := Module[{out, in, neighbors, verts},
out = Flatten@Position[M[[i], _?Positive];
in = Flatten@Position[M[[All, i], _?Positive];
neighbors = Union[out, in];
verts = Prepend[neighbors, i];
AdjacencyGraph[M[[verts, verts]], VertexLabels -> Placed["Name", Tooltip]]]
```

In[127]:=

 $\mathcal{G}_{10}$

Out[127]=



In[128]:=

GraphHub[Subgraph[G, #]] & /@ FindGraphCommunities[G]

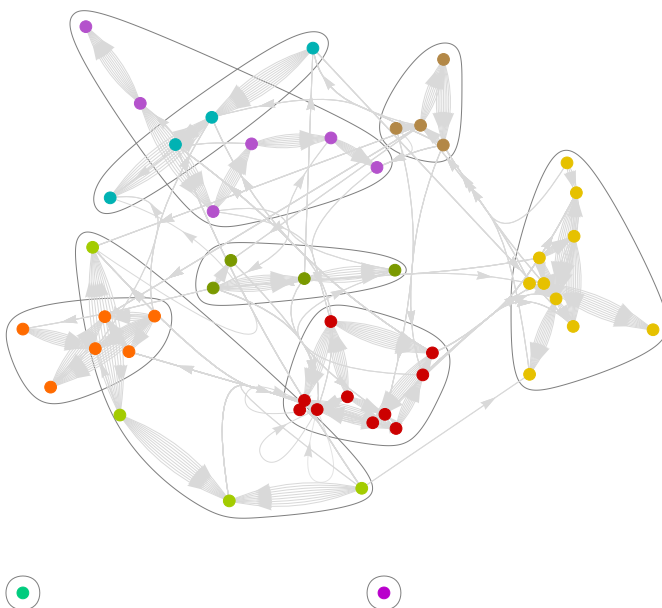
Out[128]=

 $\{\{47\}, \{3\}, \{11, 19, 29, 40\}, \{34\}, \{7, 18\}, \{12\}, \{23\}, \{37\}, \{39\}, \{50\}\}$

In[129]:=

CommunityGraphPlot[G, EdgeStyle → LightGray]

Out[129]=



In[130]:=

```
Partition[HighlightGraph[Subgraph[G, #], GraphHub[Subgraph[G, #]]] & /@
FindGraphCommunities[G], 3, 3] // GraphicsGrid
```

Out[130]=

